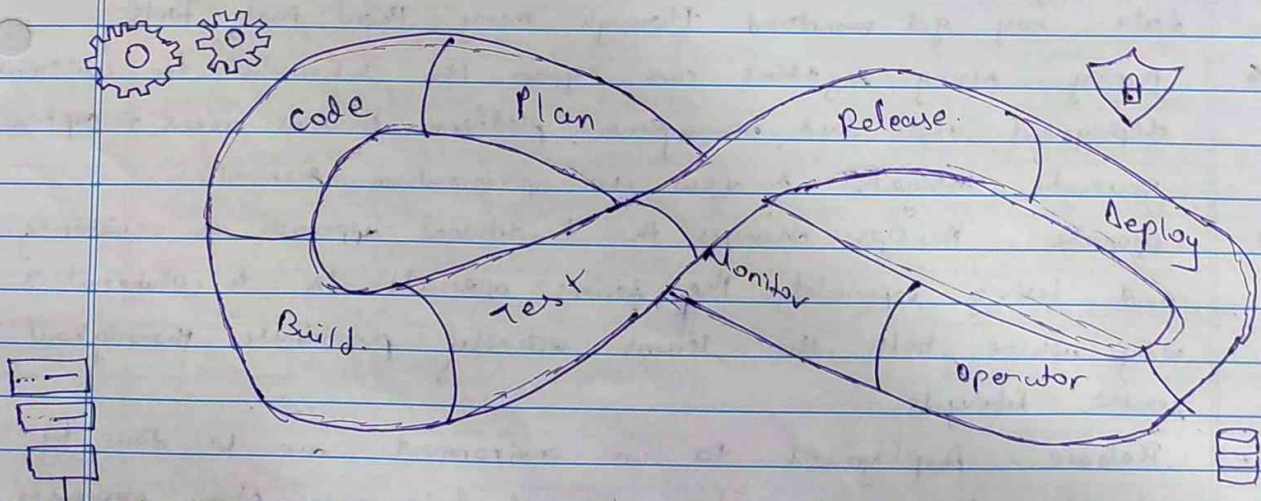


Exp. No. 1

Aim: To understand DevOps principles, practices and DevOps roles and responsibilities.

Definition:

- DevOps is the combination of two words, one is Development and the other is Operations. It is a culture to promote the development and operation process collectively.
- DevOps helps to increase organization speed to deliver applications and services. It also allows organization to serve their customers better and compete more strongly in the market.
- DevOps can also be defined as a sequence of development and IT operations with better communication and collaboration.
- DevOps has become one of the most valuable business disciplines for enterprises or organisations. With the help of DevOps, quality and speed of the application delivery has improved to a great extent.

Architecture:

1. **Build** - Without DevOps, the cost of the consumption of the resources was evaluated based on the pre-defined individual usage with fixed hardware allocation. With DevOps, the usage of cloud, sharing of resources comes into the picture and the build is dependent upon the user's need.
2. **Code** - Many good practices such as Git enables the code to be used, which ensures writing the code for business, helps to track changes and getting notified about the reason behind the difference in the actual & expected output.
3. **Test** - The application will be ready for production after testing. In the case of manual testing, it consumes more time in testing and moving the code to the output.
4. **Plan** - DevOps uses Agile methodology to plan the development. With the operations and development team in sync.
5. **Monitor** - Continuous development and monitoring is used to identify any risk of failure. Also, it helps tracking the system accurately so that the health of the application can be checked. The monitoring becomes more comfortable with services where the log data may get monitored through many third-party tools.
6. **Deploy** - Many systems can support the scheduler for automated deployment. The cloud management platform enables users to capture accurate insights & view the optimization scenarios.
7. **Operate** - DevOps changes the traditional approach of developing and testing separately. The teams operate in a collaborative way where both the teams actively participate throughout the service lifecycle.
8. **Release** - Deployment to an environment can be done by automation, it is done by manual triggering. Many processes involved in release management commonly used to do the deployment in the production environment manually to lessen

the impact on the customers.

Principles:

- Collaboration
- Data-based decision making
- Customer - Centric decision making
- Constant Improvement
- Responsibility throughout the Lifecycle
- Automation
- Failure as a Learning Opportunity.

Advantages:

- DevOps is an excellent approach for quick development and deployment of application.
- It responds faster to the market changes to improve business growth.
- DevOps escalate business profit by decreasing software delivery time and transportation costs.
- DevOps clears the descriptive process, which gives clarity on product development and delivery.
- It improves customer experience and satisfaction.
- DevOps simplifies collaboration & places all tools in the cloud for anyone to access.
- DevOps means collecting responsibility, which leads to better team engagement & productivity.

Disadvantages:

- DevOps professional or experts' developers are less available.
- Developing with DevOps is expensive.

- Adopting new DevOps technology into the industries is hard to manage in a short time.
- Lack of DevOps knowledge can be a problem in the continuous integration of automation projects.

Conclusion:

Hence, we know that DevOps has its advantages and disadvantages.